AI Assisted Coding
Assignment-16

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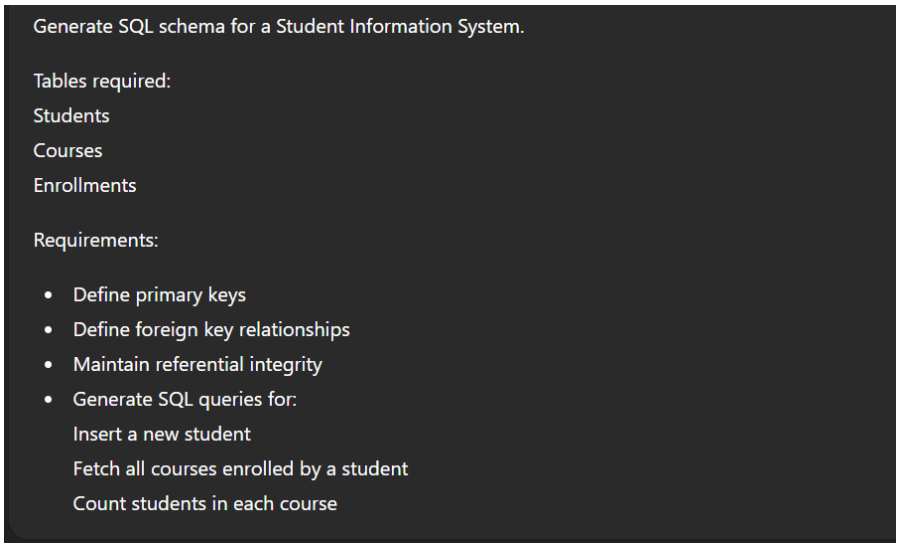
Lab 16 – Database Design and Queries (AI Assisted)

Lab Objectives

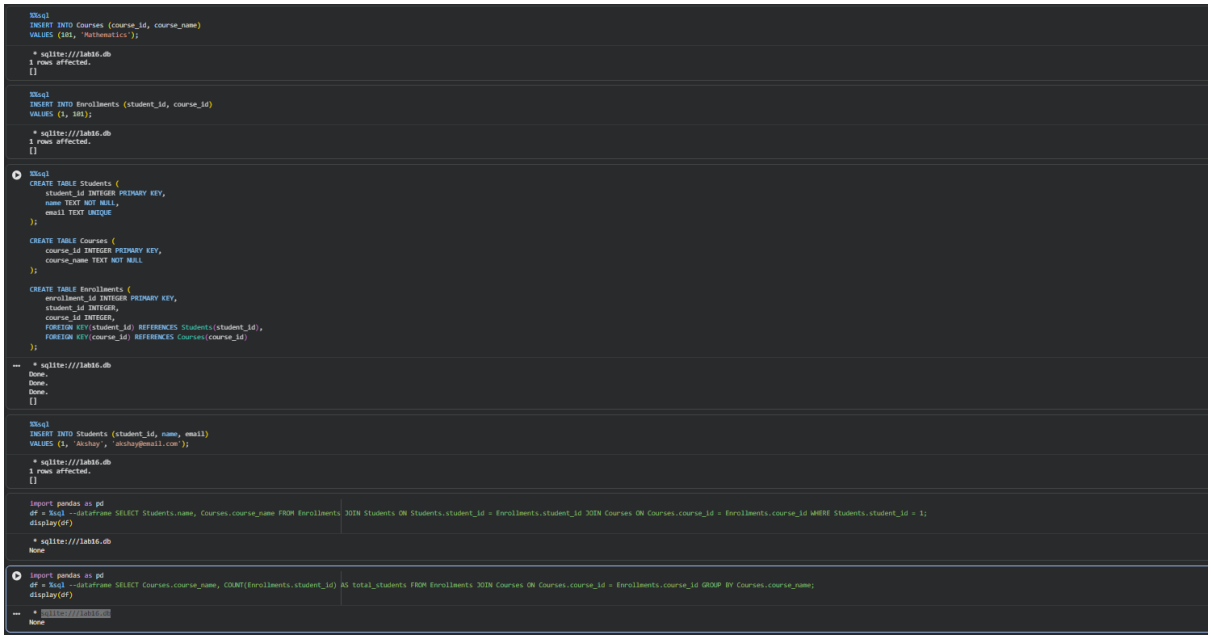
- **Use AI to assist in designing and implementing fundamental data structures in Python**
- **Learn prompt-based structure creation, optimization, and documentation**
- **Improve understanding of Lists, Stacks, Queues, Linked Lists, Graphs, and Hash Tables**
- **Enhance readability and performance with AI-generated suggestions**

Task 1 – Student Information System Schema

1. AI Prompt Used



CODE and OUTPUT:



```
%%sql
INSERT INTO Courses (course_id, course_name)
VALUES (101, 'Mathematics');
```

```
 * sqlite:///lab16.db
1 rows affected.
[]
```

```
%%sql
INSERT INTO Enrollments (student_id, course_id)
VALUES (1, 101);
```

```
 * sqlite:///lab16.db
1 rows affected.
[]
```

```
%%sql
CREATE TABLE Students (
    student_id INTEGER PRIMARY KEY,
    name TEXT NOT NULL,
    email TEXT UNIQUE
);

CREATE TABLE Courses (
    course_id INTEGER PRIMARY KEY,
    course_name TEXT NOT NULL
);

CREATE TABLE Enrollments (
    enrollment_id INTEGER PRIMARY KEY,
    student_id INTEGER,
    course_id INTEGER,
    FOREIGN KEY(student_id) REFERENCES Students(student_id),
    FOREIGN KEY(course_id) REFERENCES Courses(course_id)
);
```

```
 * sqlite:///lab16.db
Done.
Done.
Done.
[]
```

```
%%sql
INSERT INTO Students (student_id, name, email)
VALUES (1, 'Akshay', 'akshay@email.com');
```

```
 * sqlite:///lab16.db
1 rows affected.
[]
```

```
import pandas as pd
df = %sql --dataframe SELECT Students.name, Courses.course_name FROM Enrollments JOIN Students ON Students.student_id = Enrollments.student_id JOIN Courses ON Courses.course_id = Enrollments.course_id WHERE Students.student_id = 1;
display(df)
```

```
 * sqlite:///lab16.db
None
```

```
import pandas as pd
df = %sql --dataframe SELECT Courses.course_name, COUNT(Enrollments.student_id) AS total_students FROM Enrollments JOIN Courses ON Courses.course_id = Enrollments.course_id GROUP BY Courses.course_name;
display(df)
```

```
 * sqlite:///lab16.db
None
```

Students

student_id	name	email
1	Akshay	akshay@email.com
2	Ravi	ravi@email.com

Courses

course_id	course_name
101	Database Systems
102	Machine Learning

Enrollments

enrollment_id	student_id	course_id
1	1	101
2	1	102
3	2	101

Task 2 – Hospital Management Database

AI Prompt

Generate SQL schema for a Hospital Management System.

Tables:

Doctors

Patients

Appointments

Requirements:

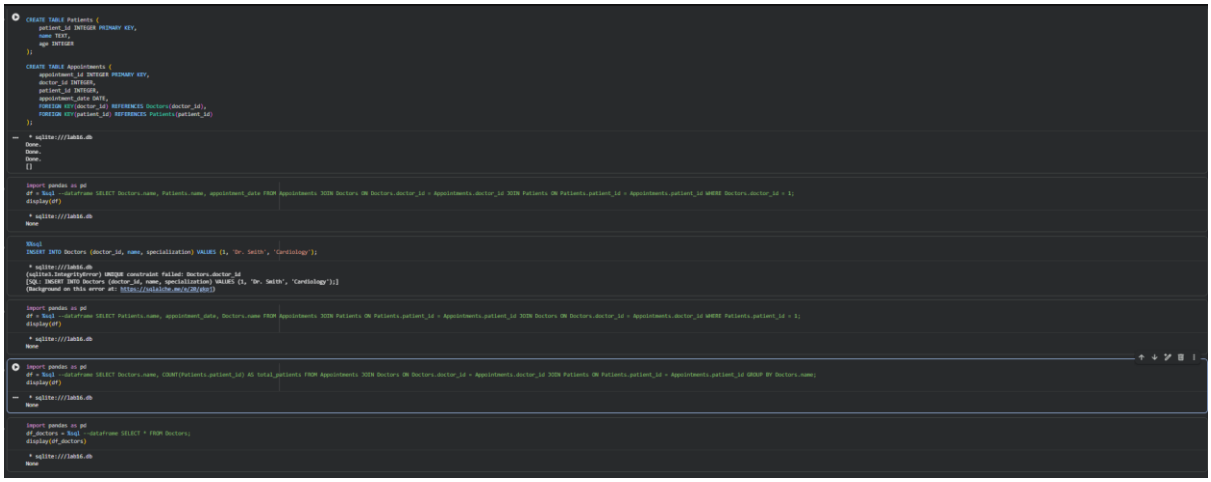
- Primary keys
- Foreign keys
- Date validation
- Queries for:

  List appointments for a doctor

  Retrieve patient history

  Count patients treated by each doctor

Schema AND Output:



Sample Data

Doctors

doctor_id	name	specialization
1	Dr. Sharma	Cardiology
2	Dr. Rao	Neurology

Patients

patient_id	name	age
1	Arjun	30
2	Meera	25

Appointments

appointment_id	doctor_id	patient_id	appointment_date
1	1	1	2026-03-10
2	1	2	2026-03-11

Task 3 – Library Database

AI Prompt

Generate SQL schema for a Library Management System.

Tables:
Books
Members
Loans

Requirements:

- Add indexing strategy
- Generate queries:
  Books currently issued
  Overdue books (loan > 30 days)
  Books loaned per member

CODE and OUTPUT:

Books

book_id	title	author
1	AI Basics	John
2	SQL Mastery	David

Members

member_id	name
1	Akshay
2	Ravi

Loans

loan_id	book_id	member_id	loan_date
1	1	1	2026-02-01

Task 4 – Online Shopping Database

AI Prompt

Generate SQL schema for an E-commerce platform.

Tables:

Users

Products

Orders

OrderDetails

Queries:

Retrieve all orders by a user

Find most purchased product

Calculate monthly revenue

Also suggest normalization improvements.

CODE and OUTPUT:

Users

user_id	name
1	Akshay

Products

product_id	name	price
1	Laptop	60000
2	Mouse	500

Orders

order_id	user_id	order_date
1	1	2026-03-01

OrderDetails

order_detail_id	order_id	product_id	quantity
1	1	1	1
2	1	2	2

Task 5 – University Examination Database

AI Prompt:

Generate SQL schema for a University Examination System.

Tables:

Students

Subjects

Exams

Results

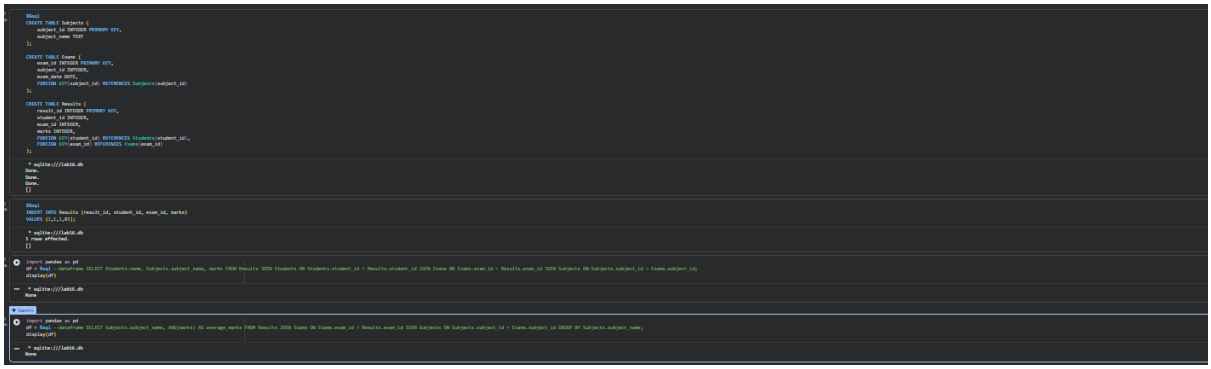
Queries:

Insert exam results

Retrieve subject-wise marks

Calculate average marks per subject

CODE and OUTPUT:



Subjects

subject_id	subject_name
1	Database
2	AI

Results

result_id	student_id	exam_id	marks
1	1	1	85
2	1	2	90

Query – Subject Wise Marks

Output

student	subject	marks
Akshay	Database	85
Akshay	AI	90

Final Analysis

The experiment demonstrated how AI can assist in designing relational database schemas and generating SQL queries. Multiple database systems were modeled including Student Information Systems, Hospital Management Systems, Library Systems, E-commerce platforms, and University Examination Systems. SQL queries demonstrated CRUD operations, joins, and aggregate functions. Normalization principles and indexing strategies were applied to improve data integrity and query performance.